MGTF-415: Collecting and Analyzing Financial Data

Instructor: Michael Reher (mreher@ucsd.edu)

Meeting Venue: 4N128

Meeting Time: Wednesdays, 11:00am (Section 1) and 2:00pm (Section 2) U.S. Pacific Time

Teaching Assistants:

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TA Zoom Office Hours: Posted to Canvas by Monday morning of the coming week.

Course Description:

This course will teach students how to obtain and process data in order to answer empirical questions in finance.

Assignments:

1. Homework: There will be three homework assignments, which will be due on 10/21, 11/4, and 11/18. Students should upload their assignments to Canvas by 11:59pm on the due date. Please save your assignment as a .pdf file and include your code in the same file as your homework assignment. Students will lose points if these formatting requirements are not met. Late assignments will receive a score of 0.

   Students will be evaluated based on the quality of their output, not the quality of their coding. However, please: write code that is computationally efficient; provide enough comments so that a reader can follow your logic, but do not comment excessively; and, as a general rule, strive to keep each line of code to under 80 characters.

   Students can work in groups, but, for logistical reasons, *each individual student should submit their own .pdf file to Canvas*, even if the file is the same as that of their group mates.

2. Final Presentation: There will be a final project which students will complete in groups of around 4 persons. Project groups will be assigned. So, like in a real company, students may not necessarily be working with their best friends. The assignment is to imagine working as an analyst in a financial services firm, and to propose a new project that the firm should undertake.

   The project topic is intentionally open-ended. The only requirements are that the project's research question would be of interest to a financial services firm and that the project's analysis is rigorously quantitative.

To provide some guidance, here are some project templates:

a. Real Estate: Imagine you work for a real estate investment trust (REIT). In class you will learn about how to study geographic variation in house prices. Propose a new geographic market in which to invest and provide evidence that investing in this geographic market is a successful business strategy.

b. Asset Management: Imagine you are working for a wealth management firm. In class you will be taught the Capital Asset Pricing Model (CAPM) as a way to understand stock returns. In this project, propose a new asset pricing model and provide evidence of its effectiveness.

c. Private Equity: Imagine you work for a data-driven venture capital firm. In class you will learn methods for obtaining data on startup companies requiring venture capital funding. Propose a new startup company in which to invest, and provide evidence that this investment will be profitable.

d. Other: Students are encouraged to think creatively, and they may of course choose a project that does not fit one of the previous three templates. The templates are here only to help students, not to be a burden.

Students will give a presentation of no longer than 8 minutes on the last class meeting day, December 2, using slides. Public speaking is an important skill, and so every student is encouraged to partake in the actual presentation. However, no points will be deducted if only part of the group speaks during the presentation. An example slide deck is available on Canvas. A rubric is on Canvas, too.

Reiterating, this project is intentionally open-ended to replicate a real-life work assignment. Students should imagine they are pitching their new project to the CEO of their firm. *It is critical that students make data-driven arguments in their presentation*. Students will be evaluated based on the quality of their analysis and the clarity of their communication. This is your opportunity to showcase your ability to collect and analyze financial data! Be creative and have fun!

Practice Exams:

I will periodically review practice exams throughout lecture. The goal of these practice exams is to make the course material more concrete. The practice exams will be posted one to two weeks before the final exam but not earlier.

Final Exam:

There will be a final exam on Wednesday, December 9. The time and location will be forthcoming. Students will take the exam in-person and on-paper. As in the homework assignments, students will be evaluated based on their understanding of fundamental principles of data analysis and finance. Students will not be evaluated based on the quality of their code, and so should not spend time memorizing syntax.

Grading:

Grades will be assigned based the final presentation (35%), final exam (35%), and homework assignments (10% each).

Code Assistance:

Students may use AI software (e.g., Claude, GPT) to assist with *coding syntax* in homework assignments or to write code for their final project. However, relying on AI for *coding logic* is highly discouraged. Doing so will limit students' understanding of the material. Most of the variation in student grades is determined by the final exam. Students who do not master core course concepts will do badly on the exam (e.g., econometric reasoning, algorithmic thinking).

Required Software:

Students may use their software of choice for completing course assignments (e.g. R, Python, Stata, etc.). Code examples in the lecture will come from Python. Students with no or minimal programming experience are strongly encouraged to use Python with a Jupyter notebook, since this will correspond most closely to the code examples in the lecture. Students can download Jupyter here: https://www.anaconda.com/distribution/

WRDS Account:

As soon as possible, students should set up a Wharton Research Data Services (WRDS) account. We will use WRDS data beginning in the fourth lecture, and so it is important that students have full access to WRDS by that point. Students can set up an account by following the directions on the front of WRDS' website here: https://wrds-www.wharton.upenn.edu/

Extra Credit:

The ability to execute trades and perform analysis using a Bloomberg Terminal is an important skill in several subsectors of the financial services industry. The course will not teach this skill explicitly, but students are strongly encouraged to acquire it by taking Bloomberg's Market Concepts Course. As an incentive, students can replace their lowest homework score with a score of 100% by taking the course, saving their certificate as a .pdf file, and uploading it to Canvas. The course can be found here: https://www.bloomberg.com/professional/product/bloomberg-market-concepts/

Additional details on setting up a Bloomberg account can be found at Rady's online portal: https://portal.rady.ucsd.edu/pages/viewpage.action?spaceKey=RTS&title=Bloomberg+Terminal+Access+from+Home

Course Outline:

(Note: This outline is provisional and subject to change)

Week 1: Trends Across the Financial Sector

Week 2: Real Estate Lecture #1

Week 3: Real Estate Lecture #2

Week 4: Asset Management Lecture #1

Week 5: Asset Management Lecture #2

Week 6: Private Equity Lecture #1

Week 7: Private Equity Lecture #2

Week 8: Environmental, Social, and Governance (ESG) Lecture

Week 9: No Class (Thanksgiving)

Week 10: Student Presentations

December 9: Final Exam